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A Problem Formulation on Sequence Tagging

Name entity recognition (NER). NER aims to predict all spans of words that represent some given classes of entity with a sentence. We adopted CoNLL03 (Sang and De Meulder, 2003), OntoNotes 5.0 (Weischedel et al., 2013) and CoNLL04 (Carreras and Màrquez, 2004). For CoNLL03 and CoNLL04, we trained our model on the standard train-develop-test split. For OntoNotes 5.0, we use the same train, develop, test split as (Xu et al., 2021). All the datasets are labeled in IOB2 format. We use sequence tagging to solve NER tasks by assigning labels marking the beginning and inside some classes of entity. The language models generate a representation for each token, and we use a linear classifier to predict the labels. We use the official scripts to evaluate the results. For the multi-task setting, we combine the training set of the three datasets for pre-training. We use different linear classifiers for each dataset while sharing the continuous prompts.

(Extractive) Question Answering (QA). Extractive QA is designed to extract the answer from the context given the context and a question. We use SQuAD (Rajpurkar et al., 2016) 1.1 and 2.0, in which each answer is within a continuous span of the context. Following tradition, we formulate the problem as sequence tagging by assigning one of the two labels: ‘start’ or ‘end’ to each token and at last selecting the span of the most confident start-end pair as the extracted answer. If the probability of the most confident pair is lower than a threshold, the model will assume the question unanswerable. For the multi-task setting, our training set for pre-training combines the training sets of SQuAD 1.1 and 2.0. When pre-training, we assume that all the questions, regardless of their origin, are possibly unanswerable.

Semantic Role Labeling (SRL). SRL assigns labels to words or phrases in a sentence that indicate their semantic roles in the sentence. We evaluate P-tuning v2 on CoNLL05 (Carreras and Màrquez, 2005) and CoNLL12 (Pradhan et al., 2012). Since a sentence can have multiple verbs, we add the target verb token to the end of each sentence to help recognize which verb is used for prediction. We classify each word with a linear classifier based on the corresponding semantic role representation. For multi-task setting, the pre-train training set is a combina-

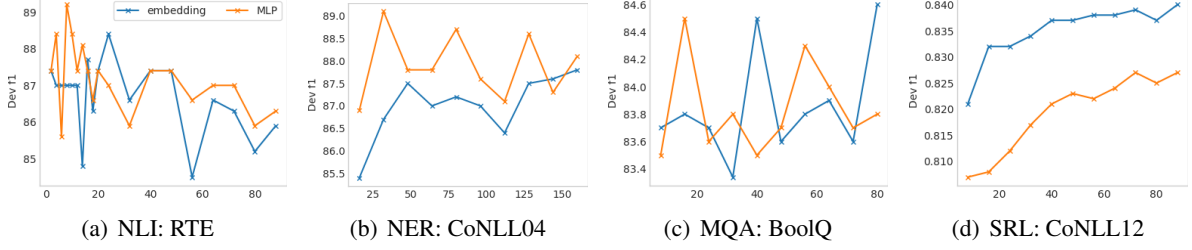


Figure 4: Ablation study on prompt length and reparameterization using RoBERTa-large. The conclusion can be very different given certain NLU task and dataset. (MQA: Multiple-choice QA)

tion of the training set of CoNLL05 (Carreras and Màrquez, 2005), CoNLL12 (Pradhan et al., 2012) and propbank-release (a common extend data used for training SRL). The multi-task training strategy is similar to NER.

B More Ablation Study

Due to the page limit, we present hyper-parameters and architecture designs ablations regarding reparameterization and prompt length in this section.

Embedding v.s. MLP reparameterization. In both prefix-tuning (Li and Liang, 2021) and P-tuning (Liu et al., 2021), authors discover the reparameterization to be useful in improving training speed, robustness and performance. However, we conduct experiments to show that the reparameterization effect is inconsistent across different NLU tasks and datasets.

As shown in Figure 4, in RTE and CoNLL04, MLP reparameterization generally indicates better performance than embedding for almost all prompt lengths. However, in BoolQ, MLP and embedding’s results are competitive; in CoNLL12, the embedding consistently outperforms MLP.

Prompt Length. Prompt length is yet another influential hyper-parameter for P-tuning v2, and its optimal value varies from task to task. From Figure 4, we observe that for simple NLU tasks, usually, a shorter prompt is enough for the best performance; for hard sequence tasks, usually, a longer prompt than 100 would be helpful.

We also discover that reparameterization has a close bond with optimal prompt length. For example, in RTE, CoNLL04, and BoolQ, MLP reparameterization achieves its optimal result earlier than embedding. This conclusion may contribute some thoughts on P-tuning’s optimization properties.